RSHS
Reduced

MATH O MAGIC

4·4

by

Karen Anderson

EDUCATIONAL DIRECTIONS PUBLICATIONS

INTRODUCTION

MATH O MAGIC 4·4 is part of a series of 12 books

MATHOMAGIC 1·2	MATHOMAGIC 4·3
MATHOMAGIC 1·3	MATHOMAGIC 4·4
MATHOMAGIC 2·3	MATHOMAGIC 5·3
MATHOMAGIC 2·4	MATHOMAGIC 5·4
MATHOMAGIC 3·3	MATHOMAGIC 6·3
MATHOMAGIC 3·4	MATHOMAGIC 6·4

These books are designed for students in years eight to ten.

1·2 and 1·3 are designed to revise the skills from earlier years and to be a platform from which high school skills are developed.

2·3, 3·3, 4·3, 5·3 and 6·3 develop skills in the SPACE, MEASUREMENT, and INFERENCE strands.

2·4, 3·4, 4·4, 5·4 and 6·4 address skills in the NUMBER and FUNCTION strands.

Stages 2, 3, 4, 5 and 6 indicate the level of difficulty, with each following in a sequential manner.

PREFACE

Often students find the necessary skill practice in Mathematics to be very boring. This series of books has been designed as a motivational tool to help the students to feel more enthusiastic about their work.

These worksheets are designed to be used in various ways –

1. MOTIVATION – Students are keen to solve the puzzle and keep asking for more.

2. INCENTIVE – As a reward for students who finish their work early.

3. SELF CHECKING
   – For students to be able to check their own work as they go along.

4. EXTRA PRACTICE
   – A fun way of doing extra practice on skills development.

5. HOMEWORK – an interesting way to get students to do more consolidation of mathematical skills in a way they enjoy.

6. REVISION – An exciting way to stimulate revision.

However you use these sheets, you will always be asked for more. The students enjoy working out the puzzles and riddles and get a great sense of satisfaction from getting them right.

Copies of this book may be obtained from:

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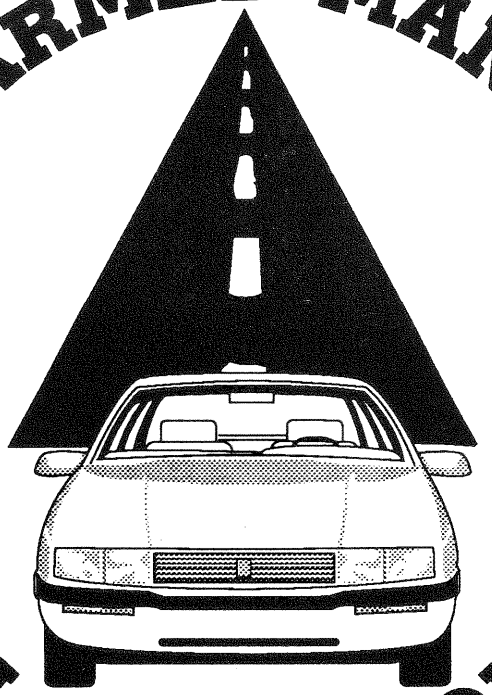
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INSTRUCTIONS:
Convert each of the fractions below to recurring decimals. Find your answer in the Code Key and notice the letter next to it. Write that letter above the number of the question to solve the puzzle below.

WHY DID THE ONE ARMED MAN CROSS THE ROAD?



CODE KEY

H – $2.\overline{63}$
D – $3.\dot{5}$
S – $0.7\dot{3}$
N – $0.\overline{692307}$
A – 1.85
C – $0.8\overline{63}$
O – $0.\overline{857142}$
G – 1.375
E – $1.\overline{71}$
P – $2.8\dot{3}$
T – $3.\overline{51}$

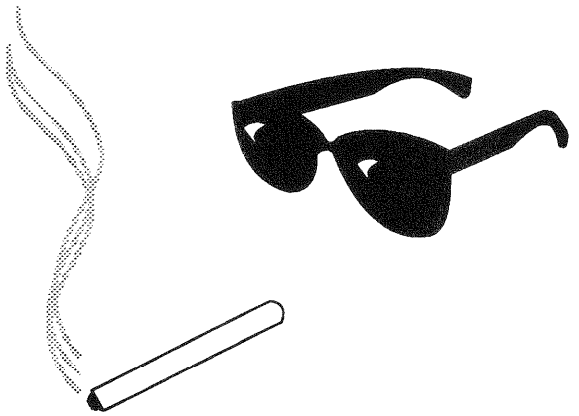
1. $\frac{9}{13}$
2. $3\frac{5}{9}$
3. $1\frac{3}{8}$
4. $\frac{19}{22}$
5. $2\frac{7}{11}$
6. $1\frac{71}{99}$
7. $\frac{6}{7}$
8. $3\frac{17}{33}$
9. $1\frac{17}{20}$
10. $\frac{11}{15}$
11. $2\frac{5}{6}$

8	7	3	6	8	8	7	8	5	6

10	6	4	7	1	2	5	9	1	2	10	5	7	11

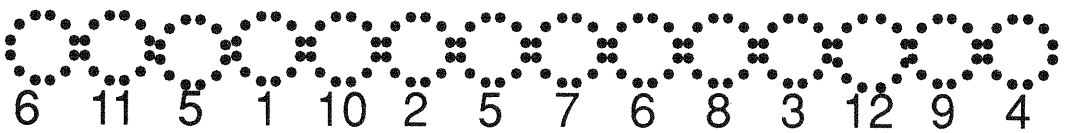
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WHAT DRINK DID THE INVISIBLE MAN GIVE HIS INVISIBLE SON?



To answer this question, rule a line between each fraction or mixed numeral in the left hand column and its decimal equivalent in the right hand column. Each line will pass through a number and a letter. Write the letter above the appropriate number in the code below.

$2\frac{2}{3}$	P	$0.\dot{8}$
$\frac{1}{6}$	1	$0.\overline{461538}$
$1\frac{6}{11}$	6	$2.\dot{6}$
$\frac{56}{99}$	7, M, R	$2.58\dot{3}$
$\frac{6}{13}$	K, E, 8, 10	$1.\overline{407}$
$2\frac{4}{5}$	12, O	$0.16\dot{6}$
$1\frac{11}{27}$	I	2.8
$\frac{13}{22}$	11, V	$0.5\overline{90}$
$\frac{8}{9}$	D	$1.\dot{5}$
$1\frac{7}{8}$	L, 9	1.875
$2\frac{7}{12}$	5, T, 3, A	$0.5\overline{6}$
$1\frac{5}{9}$		$1.\overline{54}$

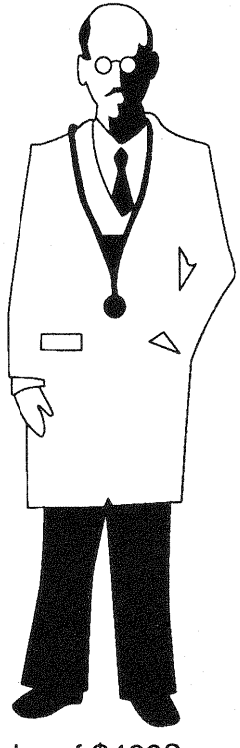


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WHAT HAPPENED TO THE PLASTIC SURGEON WHO SAT TOO CLOSE TO THE FIRE?

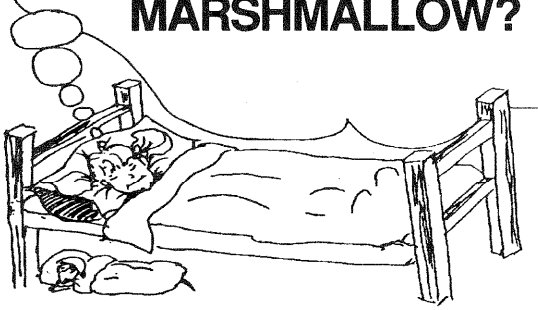
1. A bike was selling for $260. In a sale a 25% discount was given. How much will the bike cost?
2. A school has 175 students. A 16% increase is expected. How many students will there be after the increase?
3. A salesperson earned 5% commission on sales of $2 750. How much commission is that?
4. For finalizing a bill promptly, a customer was given a 2·5% discount on $268. What was the final payment on the bill?
5. At the end of last year Anna was awarded a pay rise of 6%. If she earned $605 per week last year, how much does she earn this year?
6. A certain metal expands 12·5% with heat. What length will a 12 cm metal bar expand to when heat is applied?
7. A stereo is being sold for $845. When a 40% discount is given, how much will the stereo cost?
8. David's average spelling scores improved by 20% this year. Last year his average score was 55. What was his average score this year?
9. A television is valued at $580. When it is purchased, sales tax of 10% is added. How much will a customer pay for the television?
10. Books Inc. were offering a 7·5% discount on sales over $200. How much would a school pay for an order of $460?
11. A bracelet weighs 95 g. By weight, 20% is gold. What weight of gold is contained in the bracelet?
12. A salesperson earned 5·5% commission on sales. How much would she earn for sales of $1 580?



INSTRUCTIONS:
Solve each of these problems. Look for your answers in the boxes below. Cross out each box containing a correct answer. When you finish there will be 8 boxes remaining. Rearrange the letters in numerical order to give the answer to the riddle.

3 – M $525·75	1 – M $641·30	7 – O $425·50	2 – E 10·6 cm	4 – R $261·30
5 – D 19 g	6 – T $562·34	2 – S $195	6 – A $86·90	5 – L 28
5 – E $507	8 – B 66	8 – D 39	1 – H $33·50	2 – N 203
2 – G $137·50	4 – E $96·20	1 – L 13·5 cm	3 – S $638	7 – E $142·75

WHAT HAPPENED TO THE GIRL WHO DREAMED SHE WAS EATING A MARSHMALLOW?



Determine what percentage of the original amount will be given by the opposite.



Look for your answers in the code below. Write the letter of each question above its answer.

K	an increase of 25%
W	an increase of 37%
D	a decrease of 12%
O	an increase of 8·5%
L	a decrease of 15%
U	a decrease of 6%
H	a decrease of 23%
R	an increase of 4%
E	an increase of 7·5%
A	a decrease of 52%
N	an increase of 13·25%
S	a decrease of 22·5%
I	a decrease of 33%
P	an increase of 17%

137	77	107·5	113·25	77·5	77	107·5	137	108·5	125	107·5
94	117	77	107·5	104	117	67	85	85	108·5	137
88	67	77·5	48	117	117	107·5	48	104	107·5	88

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WHAT DO CANNIBALS HAVE FOR BREAKFAST?



INSTRUCTIONS:
Find the percentage increase or decrease in the following questions. Look for your answers in the boxes below. Cross out each box containing a correct answer. When you finish, write the letters in the remaining boxes in order from left to right.

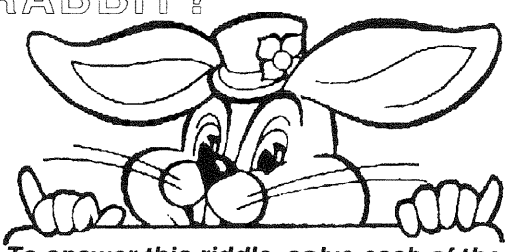
1. A doctor's fee increased from $30 to $36. What percentage increase is this?
2. Following a diet, a person's weight reduced from 120 kg to 75 kg. What percentage decrease from the original weight is this?
3. A television originally priced at $750 was on sale for $660. What percentage discount is this?
4. Attendances at the Easter Show on Friday were 15 860. On Saturday, 18 239 people attended. From Friday to Saturday what was the percentage increase?
5. When Janet began weight training her weight increased from 50 kg to 64 kg in twelve months. What percentage increase is this?
6. During a winter sale, coats originally priced at $150 were on sale for $124·50. What percentage discount is this?
7. A school's population of 150 was expected to increase to 171 during the following year. What percentage increase is that?
8. A skateboard was advertised for $65 before Christmas. During a post-Christmas sale the same skateboard was on sale for $49·40. What percentage discount is this?
9. Following an advertising campaign, a shop's icecream sales rose from $18 500 per week to $24 605 per week. What percentage increase is this?
10. At the end of last year, Darren's salary increased from $22 400 to $24 192. What percentage raise did Darren receive?
11. A computer normally priced at $2 144 was advertised in the newspaper for $1 876. What percentage discount was being offered?

BA	SH	AN	KE	TS
21%	8%	20%	7·5%	17%
DB	**KL**	**EI**	**CK**	**NG**
31%	14%	19%	24%	29%
OA	**ET**	**SO**	**DA**	**NT**
28%	12·5%	36%	37·5%	5%
OA	**NL**	**KE**	**ST**	**MS**
10%	12%	33%	30%	15%

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WHAT DO YOU GET IF YOU CROSS A GARDEN SPRINKLER WITH A RABBIT?



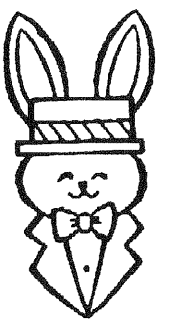


To answer this riddle, solve each of the problems. Look for your answers in the Code Key. Notice the letter next to each answer. Write this letter above the number of the question in the puzzle at the bottom of the page.

1. After a 10% discount, a stereo was on sale for $315. What was the price of the stereo before the discount?
2. Following a 15% pay rise, Robyn is now earning $644 per week. What was her wage prior to the rise?
3. A newspaper's circulation increased 12% to be selling 284 480 copies per week. How many papers were being sold before the increase?
4. A pair of basketball boots was selling for $132, after a discount of 20%. What was the price of the boots before the discount?
5. On the 1st January, doctors' fees rose 8%. Patients now pay $35·10 per consultation. How much did patients pay before the price rise?
6. When a particular maths book was discounted by 12·5%, its selling price became $4·55. What was the book's price before the discount?
7. A large mirror was discounted by 7·5% to be selling now for $77·70. What was the price of the mirror before the discount?
8. This year 136 728 voters voted in the local council elections, an increase of 5·5% more than the number of people who voted last year. Determine how many people voted last year.
9. Final day tickets for a tennis tournament were selling for $34·85, a discount of 18%. What was the full price of a ticket, before the discount?

CODE KEY

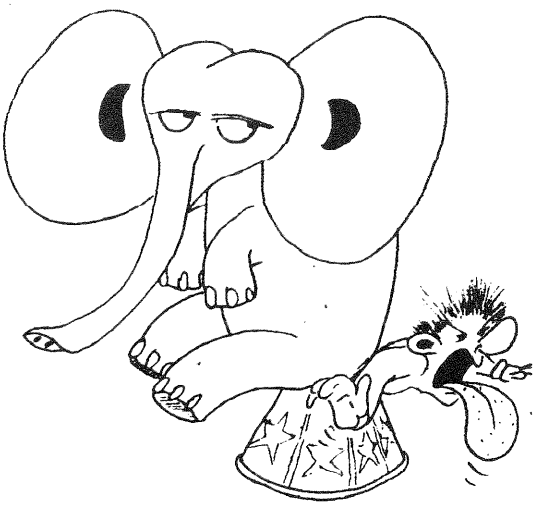
$32·50	R
$560	Y
$165	H
129 600	S
$350	P
$42·50	R
$5·20	A
254 000	E
$84	A



4	6	9	3	8	1	5	7	2

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WHAT DID RAY BECOME WHEN THE ELEPHANT STEPPED ON HIM?



Use your calculator to find the answers to the following, correct to 2 decimal places. Look for your answers at the bottom of the page. Cross out each box containing a correct answer. You will be left with five boxes. Write the remaining letters in order from left to right.

$\sqrt{60}$	$\sqrt[3]{10}$	$\sqrt[3]{50}$
$\sqrt{120}$	$\sqrt{300}$	$\sqrt{22}$
$\sqrt[3]{28}$	$\sqrt[3]{90}$	$\sqrt[3]{158}$
$\sqrt[3]{220}$	$\sqrt{5}$	$\sqrt{39}$
$\sqrt{95}$	$\sqrt[3]{68}$	$\sqrt{183}$
$\sqrt[3]{195}$	$\sqrt[3]{375}$	$\sqrt{275}$
$\sqrt{500}$	$\sqrt[3]{550}$	

N	S	E	L	U	M	E	P	L	X	N	R	O	E	D	L	A	N	D	S	E	K	Y	N	S
10·95	2·15	7·34	6·24	3·04	4·08	22·36	6·04	7·75	2·05	9·75	8·11	2·24	8·19	3·68	5·41	13·28	7·21	17·32	13·53	16·58	4·69	12·75	4·48	5·80

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*Without using your calculator, choose the **best estimate** for each problem, from the row next to it. Circle the letter next to it. Write that letter at the bottom of the page above the number of that question.*

HOW DID THE NERVOUS CARPENTER BREAK HIS TEETH?



1.	$\sqrt{86}$	D	8·5	T	9·5	S	10
2.	$\sqrt{450}$	N	22	P	55	L	120
3.	$\sqrt[3]{15}$	I	2·5	E	5	A	3·5
4.	$\sqrt[3]{100}$	W	33	T	5·5	S	4·5
5.	$\sqrt[3]{950}$	O	300	I	9·5	E	32
6.	$\sqrt{150}$	M	10·5	B	75	H	12·5
7.	$\sqrt[3]{2\,000}$	I	12	A	600	U	20
8.	$\sqrt{0{\cdot}5}$	N	0·5	S	0·25	H	0·7
9.	$\sqrt{3}$	O	1·9	E	1·7	A	1·5
10.	$\sqrt[3]{0{\cdot}5}$	R	0·1	S	0·8	N	0·25
11.	$\sqrt[3]{50}$	L	3·5	R	17	K	7·5
12.	$\sqrt[3]{4\,000}$	B	1 000	D	130	B	15
13.	$\sqrt{1\,000}$	U	500	E	100	A	32

6	9	12	5	1	8	3	10	2	13	7	11	4	

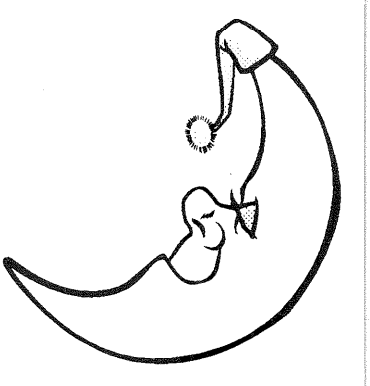
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WHY DID THE WOMAN SLEEP ON THE CHANDELIER?

INSTRUCTIONS:
For each of the following, decide whether the the two quantities are directly proportional, inversely proportional or not proportional at all. Circle the number – letter combination in the appropriate column. Then write the letter above the number in the code at the bottom.

	PROPORTIONALITY		
	Direct	**Inverse**	**None**
the number of compact discs and the total cost	**5 – R**	**10 – A**	**14 – Y**
the number of lollies you have and the time taken to eat them	**11 – K**	**14 – D**	**2 – H**
the rate at which grass grows and the number of times it is mowed	**6 – G**	**13 – H**	**12 – B**
the speed of a car and the time taken to arrive at a destination	**9 – N**	**8 – W**	**1 – T**
the time taken to deliver and the number of people delivering a wad of pamphlets	**4 – T**	**14 – A**	**3 – N**
the number of dollar coins in a pile and the height	**4 – P**	**11 – M**	**6 – N**
a person's maths ability and the distance they live from school	**9 – A**	**7 – P**	**13 – T**
the constant spped of a car and the distance covered	**1 – S**	**2 – G**	**4 – L**
the mass of people in grams and their mass in kilograms	**9 – I**	**3 – D**	**8 – R**
the length of time spent training and an athlete's running speed	**7 – R**	**13 – S**	**6 – L**
the number of people and the time spent by each to staple 500 newsletters	**8 – H**	**10 – E**	**12 – G**
the length of paces and the number of paces taken in 100 m	**12 – S**	**11 – U**	**1 – R**
the length of people's feet and their weight	**14 – E**	**10 – O**	**3 – C**
the number of two hour videos and the time taken to watch them	**7 – G**	**2 – P**	**14 – U**

12	10	3	14	11	1	10	1	2	10	8	14	1
14	6	9	7	2	13	1	6	10	10	4	10	5



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WHAT SCHOOL DO YOU HAVE TO DROP OUT OF BEFORE YOU GRADUATE?

Solve the following problems. Look for your answers in the CODE KEY. Notice the letter next to each answer. Write that letter above the question number in the code below.

1. If 5 pies cost $11·20, how much would 8 pies cost?
2. An ore truck can deposit 12·9 tonnes in 6 trips. How much can be deposited in 10 trips?
3. Eleven cases of apples cost $68·75. What is the cost of 7 cases?
4. If it costs $945 to rent a house for 7 weeks, how much would it cost to rent for 3 weeks?
5. A 12 m length of pipe weighs 31·2 kg. What is the weight of 8 m of the same pipe?
6. A 9 storey building is 28·8 m high. How tall would a 5 storey building be?
7. If 5 m of cord costs $7, how much would 2 m of the cord cost?
8. Three phone books weigh 5·1kg. How much would 7 phone books weigh?
9. Four calculators were selling for $75·60. How much would you pay for 5 calculators?
10. When a 10% discount is given, $58·40 is taken off the price of a television. What price would be deducted if an 18% discount was given?
11. When a water tank was three-quarters full, it held 732 litres. How much would the tank hold if it was completely full?

CODE KEY

11·9 kg	A
$405	L
21·5 t	R
16 m	P
976 L	E
94·50	O
$17·92	T
$2·80	S
20·8 kg	H
$43·75	C
$105·12	U

6	8	2	8	3	5	10	1	11	7	3	5	9	9	4

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WHY DID THE LITTLE BOY DANCE ON THE JAR OF JAM?

INSTRUCTIONS:
Simplify the expressions below. Look for your answers in the column on the right. Circle the correct answer from the three choices. Notice the letter next to each answer. Write this letter above the number of the question at the bottom of the page.



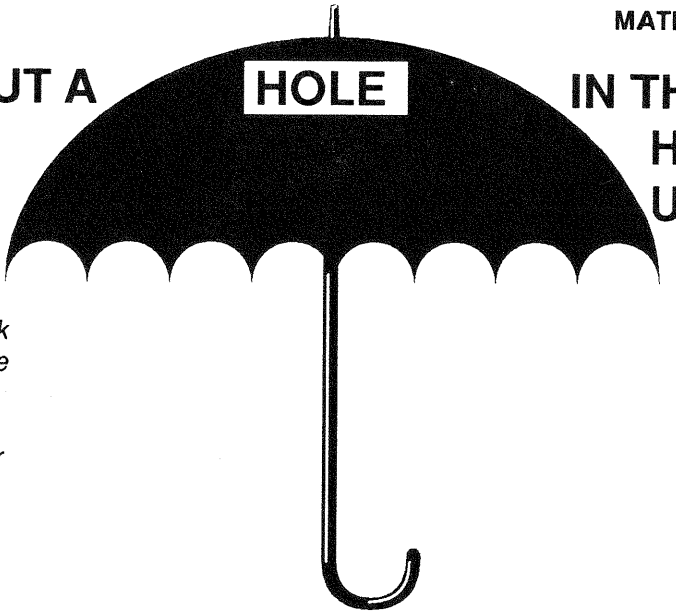
1.	$6^2 \cdot 6^9$	6^{18}	**D**	6^7	**T**	6^{11}	**U**
2.	$4^3 \cdot 4^2 \cdot 4^4$	4^9	**O**	4^{10}	**E**	4^{24}	**R**
3.	$5^3 \cdot 5^{-2} \cdot 5^4$	5^{-24}	**N**	5^{10}	**T**	5^5	**A**
4.	$10^3 \cdot 10^{-5} \cdot 10^6$	10^{13}	**Y**	10^4	**W**	10^{-4}	**L**
5.	$2^7 \cdot 2^{-1} \cdot 2^{-10}$	2^4	**G**	2^{-70}	**M**	2^{-4}	**B**
6.	$x^3 \cdot x^4 \cdot x^6$	x^{18}	**H**	x^{13}	**L**	x^{22}	**N**
7.	$y^4 \cdot y^{-2} \cdot y^5$	y^7	**N**	y^{-3}	**K**	y^{-40}	**B**
8.	$t^3 \cdot t^3 \cdot t^{-1}$	t^5	**C**	t^8	**D**	t^{-6}	**P**
9.	$m^{-3} \cdot m^4 \cdot m^{-2}$	m^{24}	**T**	m^{-1}	**S**	m^{-24}	**M**
10.	$k^6 \cdot k^{-7} \cdot k^{-2}$	k^{-3}	**T**	k^3	**N**	k^{15}	**H**
11.	$2x^3 \cdot x^5$	$4x^8$	**M**	$32x^{15}$	**R**	$2x^8$	**D**
12.	$y^5 \cdot 3y^2 \cdot 2y$	$5y^8$	**N**	$6y^8$	**H**	$5y^7$	**T**
13.	$m^2 \cdot 4m^3 \cdot 3m$	$12m^6$	**P**	$7m^5$	**S**	$12m^5$	**D**
14.	$2x^{-4} \cdot 3x^{-5}$	$6x^{-9}$	**I**	$5x^{20}$	**A**	$6x^{20}$	**O**
15.	$5k^2 \cdot 2k^{-4} \cdot 3k$	$30k^{-2}$	**L**	$30k^{-1}$	**E**	$10k^{-4}$	**A**

	5	15	8	3	1	9	15	10	12	15	6	14	11	
9	3	14	11	10	4	14	9	10	10	2	2	13	15	7



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WHY DID THE GIRL CUT A HOLE IN THE TOP OF HER UMBRELLA?



INSTRUCTIONS:
Simplify the expressions below. Look for your answers in the column on the right. Circle the correct answer from the three choices. Notice the letter next to each answer. Write this letter above the number of the question at the bottom of the page.

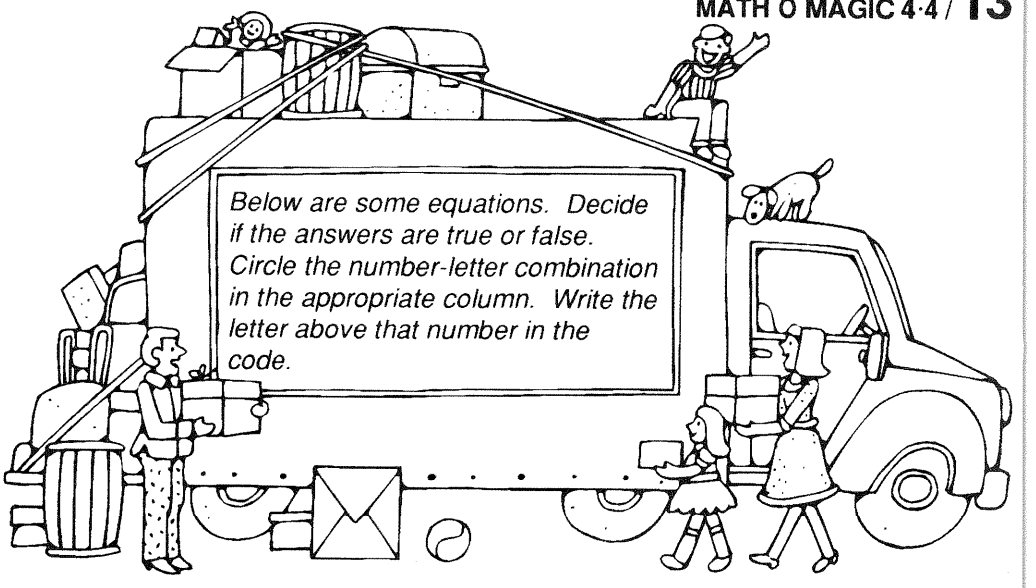
1.	$(3^4)^2$	3^6	**E**	3^8	**O**	9^4	**N**
2.	$(2^6)^3$	2^{18}	**R**	2^9	**T**	8^6	**K**
3.	$(5^4)^2$	5^6	**D**	5^8	**S**	25^4	**F**
4.	$(7^5)^4$	7^9	**D**	49^7	**A**	7^{20}	**U**
5.	$(x^3)^7$	x^4	**M**	x^{10}	**F**	x^{21}	**G**
6.	$(m^5)^3$	m^{15}	**P**	m^8	**R**	m^2	**Y**
7.	$(y^4)^3$	y	**E**	y^7	**O**	y^{12}	**I**
8.	$(k^4)^6$	k^{10}	**G**	k^{24}	**D**	k^{-2}	**B**
9.	$(x^{-3})^2$	x^{-6}	**H**	x^{-5}	**N**	x^{-1}	**S**
10.	$(t^4)^{-3}$	t^{-7}	**T**	t	**E**	t^{-12}	**A**
11.	$(y^{-4})^{-2}$	y^8	**C**	y^{-6}	**L**	y^6	**R**
12.	$(2m^2)^3$	$8m^5$	**G**	$8m^6$	**N**	$2m^6$	**T**
13.	$(3x^{-5})^2$	$3x^{-10}$	**H**	$9x^{-10}$	**W**	$9x^{-3}$	**Y**
14.	$(4y^{-3})^3$	$64y$	**B**	$4y^0$	**M**	$64y^{-9}$	**L**
15.	$(2y)^2 \cdot (3y^5)^3$	$108y^{10}$	**B**	$108y^{17}$	**T**	$6y^{17}$	**C**
16.	$(3k^{-2})^2 \cdot (2k^4)^3$	$72k^8$	**E**	$6k^7$	**A**	$72k^{-8}$	**N**

3	1	3	9	16	11	1	4	14	8	3	16	16
13	9	16	12	7	15	3	15	1	6	6	16	8
2	10	7	12	7	12	5						



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WHAT PIECE OF FURNITURE DID THE MAN OPEN AND TWENTY PEOPLE FELL OUT?



Below are some equations. Decide if the answers are true or false. Circle the number-letter combination in the appropriate column. Write the letter above that number in the code.

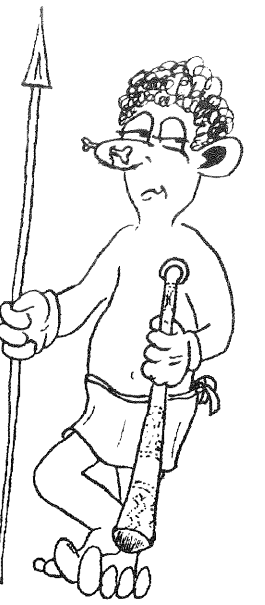
	TRUE	FALSE
$\frac{5^9}{5^3} = 5^6$	9 – P	4 – K
$\frac{3^{12}}{3^4} = 3^3$	6 – E	2 – N
$\frac{2^7}{2} = 2^7$	3 – A	7 – O
$4^7 \div 4^9 = 4^{-2}$	12 – G	9 – L
$10^2 \div 10^8 = 10^{-6}$	1 – R	2 – D
$x^{14} \div x^2 = x^7$	12 – T	10 – M
$\frac{m^9}{m^3} = m^3$	8 – M	4 – S
$\frac{k^2}{k^{-3}} = k^5$	6 – U	7 – G
$\frac{6x^{10}}{2x^3} = 4x^7$	10 – Y	11 – E
$\frac{12\,m^4}{3\,m} = 4m^3$	8 – B	5 – E
$\frac{16k^3}{2k^5} = 8k^{-2}$	3 – A	1 – T
$\frac{20x^5}{4x^{-2}} = 16x^3$	11 – A	5 – I

3	10	5	4	4	5	2	12	9	11	1	4	7	2	4

8	6	1	11	3	6

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What game do cannibal children like to play?



Simplify the left hand column below by multiplying the factors. Rule a line to connect the factors in the left hand column with the simplified version in the right hand column. Each line will pass through a number and a letter. Write the letter above the number in the code below.

$(x + 5)(x + 3)$ ◆		◆ $x^2 - 16$
$(x - 7)(x + 2)$ ◆		◆ $x^2 + 8x + 15$
$(x - 9)(x - 1)$ ◆		◆ $x^2 - 9x + 20$
$(x - 3)(x + 4)$ ◆		◆ $x^2 - 10x + 9$
$(x - 4)(x - 5)$ ◆		◆ $x^2 - 1$
$(x + 6)(x + 5)$ ◆		◆ $x^2 - 6x - 27$
$(x - 3)(x + 6)$ ◆		◆ $x^2 - 5x - 14$
$(x - 4)(x + 4)$ ◆		◆ $x^2 - 11x + 30$
$(x + 9)(x + 3)$ ◆		◆ $x^2 + 4x - 12$
$(x - 2)(x + 4)$ ◆		◆ $x^2 + 3x - 18$
$(x - 5)(x - 6)$ ◆		◆ $x^2 + x - 12$
$(x + 7)(x - 5)$ ◆		◆ $x^2 + 12x + 27$
$(x - 9)(x + 3)$ ◆		◆ $x^2 + 2x - 35$
$(x - 1)(x + 1)$ ◆		◆ $x^2 + 9x + 20$
$(x + 6)(x - 2)$ ◆		◆ $x^2 + 2x - 8$
$(x + 4)(x + 5)$ ◆		◆ $x^2 + 11x + 30$

7	16	11	4	13	8	3	14	9	15

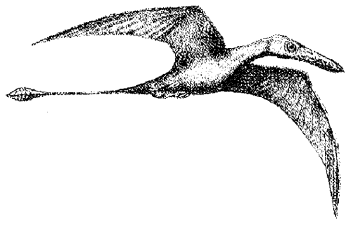
1	5	10	6	12	2



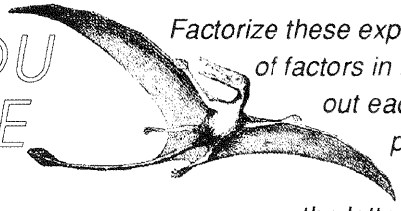


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WHAT DO YOU GET IF YOU CROSS A SHARK WITH THE LOCH NESS MONSTER?



Factorize these expressions. Look for the pair of factors in the boxes below and cross out each box containing a correct pair. When you finish there will be 4 boxes left. Write the letters in order from left to right.

$x^2 - x - 20$

$x^2 + 2x + 1$

$x^2 - 11x + 30$

$x^2 - 7x + 12$

$x^2 + 12x + 32$

$x^2 - 11x + 18$

$x^2 + 12x + 35$

$x^2 - 5x + 6$

$x^2 + 2x - 35$

$x^2 + 6x + 9$

$x^2 + x - 30$

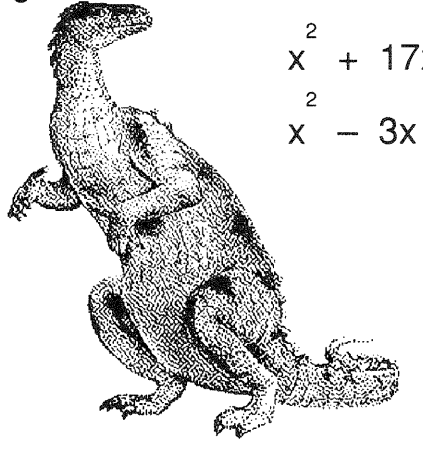
$x^2 + 8x + 12$

$x^2 - 14x + 45$

$x^2 - 7x - 8$

$x^2 + 17x + 72$

$x^2 - 3x - 40$



(x + 3)(x + 3) **SH**	(x + 4)(x + 8) **AN**	(x + 5)(x + 1) **LO**	(x – 9)(x – 2) **WS**	(x + 1)(x + 1) **CH**
(x – 5)(x + 6) **EW**	(x – 3)(x – 5) **CH**	(x + 4)(x – 5) **EN**	(x + 9)(x + 8) **SA**	(x – 8)(x +1) **NC**
(x – 2)(x – 3) **DE**	(x + 7)(x + 5) **AL**	(x – 3)(x – 4) **LK**	(x + 2)(x + 10) **JA**	(x + 6)(x + 2) **YS**
(x – 4)(x – 5) **WS**	(x + 7)(x – 5) **ME**	(x + 5)(x – 8) **OT**	(x – 9)(x – 5) **NE**	(x – 5)(x – 6) **TO**



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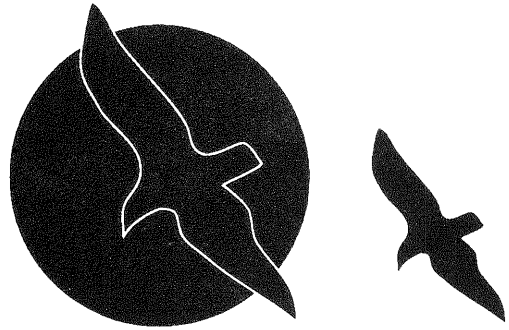
WHEN YOU CROSS A WOODPECKER WITH A CARRIER PIGEON YOU GET A BIRD...

Factorize the expressions below and look for the factors in the Code Key. Write the letter next to each answer above the number of each question in the box provided.

2	12	5	2	6	14	11	13	6	9

11	14	2	12	7	15	11	11	4	5	9	3	2	

15	7	16	3	10	7	4	9	1	7	9	9	5	8	7	9



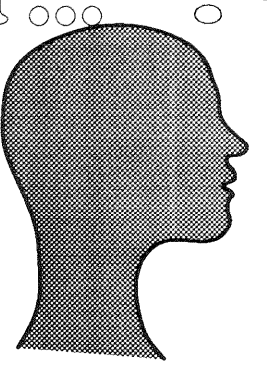
1. $x^2 - 9$
2. $x^2 - 64$
3. $4x^2 - 49$
4. $9x^2 - 25$
5. $25x^2 - 81$
6. $49x^2 - 4y^2$
7. $x^2 - y^2$
8. $x^2 - 9y^2$
9. $16x^2 - 25y^2$
10. $25x^2y^2 - 1$
11. $64x^2y^2 - 9z^2$
12. $1 - 49x^2y^2$
13. $4x^2 - y^2$
14. $1 - 81x^2y^2$
15. $9x^2 - y^2$
16. $9y^2 - 1$

CODE KEY

V	$(5xy - 1)(5xy + 1)$	**A**	$(5x - 9)(5x + 9)$
R	$(3x - 5)(3x + 5)$	**L**	$(3y - 1)(3y + 1)$
G	$(x - 3y)(x + 3y)$	**D**	$(3x - y)(3x + y)$
M	$(x - 3)(x + 3)$	**S**	$(4x - 5y)(4x + 5y)$
T	$(x - 8)(x + 8)$	**N**	$(1 - 9xy)(1 + 9xy)$
K	$(7x - 2y)(7x + 2y)$	**C**	$(2x - y)(2x + y)$
O	$(8xy - 3z)(8xy + 3z)$	**H**	$(1 - 7xy)(1 + 7xy)$
E	$(x - y)(x + y)$	**I**	$(2x - 7)(2x + 7)$

IF YOU CROSSED AN OCTOPUS WITH A MINK YOU'D GET...

To complete this sentence, simplify each of the expressions in the lefthand column by combining like terms. Find the simplified form in the right hand column. Rule a line to match the pair. Each line will pass through a number and a letter. Write the letter above the number at the bottom of the page.

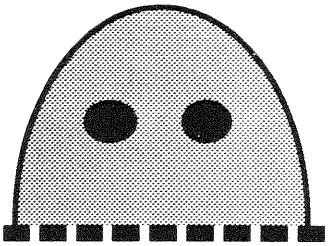


$2(3x + 4)$	✳	$10x - 15$
$4x + 3(x - 2)$	✳	$40 - 10x$
$10 + 5(6 - 2x)$	✳	$5x - 13$
$x + 2x + 2(x + 5)$	✳	$3x + 2$
$4(6 - x)$	✳	$14x - 8$
$3x - 2(2x + 3)$	✳	$6x + 8$
$5(2x - 3)$	✳	$-3x - 6$
$7 + 2x + 6 - 5x$	✳	$18x - 12$
$6 - 3(x + 4)$	✳	$7x - 6$
$5x + 2x - 3 - x$	✳	$11 - 2x$
$4(5 - 3x)$	✳	$24 - 4x$
$2(3 - 2x) + 5 + 2x$	✳	$-x - 6$
$2 - 8x + 5x + 6x$	✳	$7x - 2$
$3(4x - 1) + 2x - 5$	✳	$20 - 12x$
$6(3x - 2)$	✳	$5x + 10$
$7 + 4x - 9 + 2x + x$	✳	$6x - 3$
$3(5 - 2x) + 4x + 7(x - 4)$	✳	$13 - 3x$

C 16 4 2 9 12 V F I U 15 T M 8 N 17 14 H 7 W 10 11 13 L 5 O 1 Y 3 R A 6 E S

6	4	9	12	16	1	6	10	8	2	10	3	
10	1	1	14	6	17	15	7	11	13	13	5	13	7

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WHAT HAPPENED WHEN THE KING'S MEN PLAYED A JOKE ON HUMPTY DUMPTY?



6 9 11 4 7 3 1 10 5 8 2

For each of the following tables of patterns, decide whether the situation is LINEAR, QUADRATIC or EXPONENTIAL. Circle the number-letter combination in the appropriate columns on the right. Then write the letter above the number in the code above.

x	1	2	3	4	5
y	3	4	6	9	13

x	1	2	3	4	5
y	4	7	10	13	16

x	1	2	3	4	5
y	1	2	4	8	16

x	1	2	3	4	5
y	2	6	18	54	162

x	1	2	3	4	5
y	5	9	13	17	21

x	1	2	3	4	5
y	3	5	9	15	23

x	1	2	3	4	5
y	64	32	16	8	4

x	1	2	3	4	5
y	4	7	12	19	28

x	1	2	3	4	5
y	3	6	12	24	48

x	1	2	3	4	5
y	6	10	14	18	22

x	1	2	3	4	5
y	5	8	14	23	35

LINEAR	QUADRATIC	EXPONENTIAL
6 – M	11 – F	8 – A
3 – L	10 – E	4 – R
11 – G	1 – T	7 – L
10 – A	3 – N	6 – H
2 – T	7 – B	9 – A
9 – O	4 – E	5 – L
2 – S	3 – C	10 – O
7 – S	8 – I	11 – D
1 – W	6 – R	5 – R
9 – E	8 – U	2 – M
5 – T	1 – F	4 – A

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WHAT WOULD YOU GET IF YOU SWALLOWED URANIUM?

Solve each of the following equations and find your answers in the spaces below.
Write the letter of each question above its solution.

M $2x + 3 = 6x - 2 - 3x$

A $10 - 3x = 2 - 5x$

H $12 - 2(2x + 1) = 5x + 1$

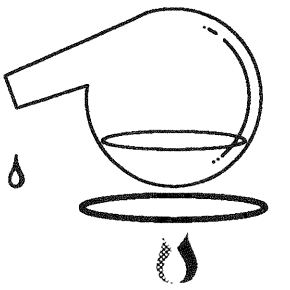
O $10 + 4x = 2x - 4$

E $3(x + 4) = 2(2x - 5) + 13$

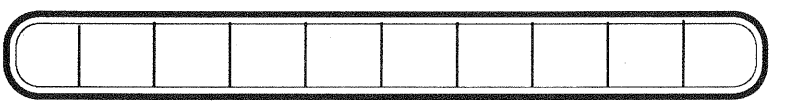
I $3x - 2(x + 4) = x + 4x$

T $5x - 5 = 2(3x - 1) - 2(x + 3) + 7$

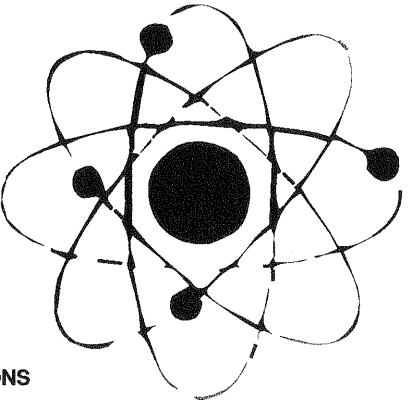
C $16 - 2x = 4x - 20 - 3x + 3$



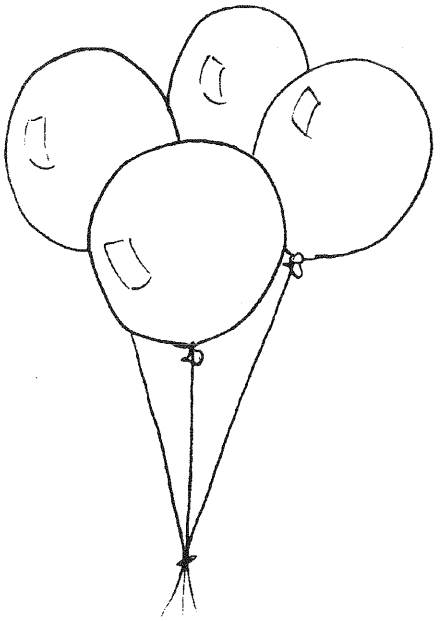
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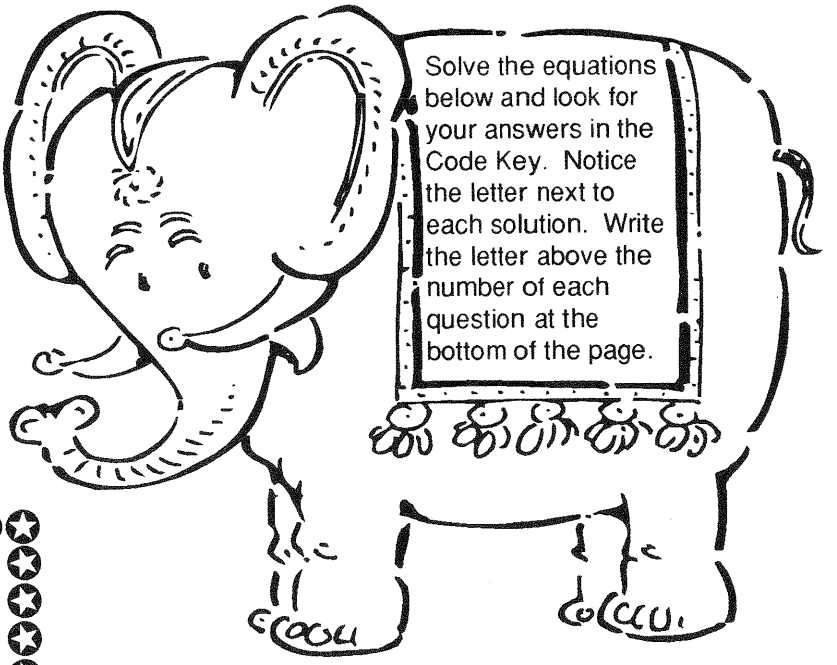
-4 4 -7 5 -2 11 -4 11 1 9



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WHEN THE SWORD SWALLOWER WENT ON A DIET, WHAT DID HE HAVE TO EAT?



Solve the equations below and look for your answers in the Code Key. Notice the letter next to each solution. Write the letter above the number of each question at the bottom of the page.

1. $4(2x + 7) = 92$
2. $9x + 25 + 8x = -26$
3. $3(2x + 1) + 5 = 32$
4. $15 - (2x + 7) = 16$
5. $8x + 3(2 - 3x) = 4$
6. $-3(x + 5) = 2(x + 5)$
7. $3(x - 4) = x + 12$
8. $5(1 + x) = 6(x + 2)$

CODE KEY

4	:	S
2	:	A
12	:	N
8	:	D
− 4	:	L
− 7	:	E
− 3	:	I
− 5	:	P

6	2	7	3	5	7	1	7	8	8	1	4	8	3

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WHAT HAPPENED AT THE BADLY ORGANIZED MILKING CONTEST?

INSTRUCTIONS:
Solve each of the following equations. Find your solutions in the boxes below. Cross out each box containing a correct answer. When you finish, write the remaining letters in order from left to right and you will have the answer to the riddle.



$2x + 4(x + 1) = 2(4x + 1) - x - 1$

$3(2 - x) + 2x - 12 = 5x - 2(2x + 1)$

$7 + 3x - 5(x - 2) = 4(2 - 3x) - 1$

$-3(x + 4) + 2x = 4 - 4x + 2$

$8 + 2(3x - 1) = 4x + x + 1$

$-2(5 + 2x) = 4(5 - 3x) + 5x$

$3 - 3(2x - 5) - 10 = -7x + 2(x + 8)$

$4(5 - x) - 6 = 6x + 5 + 8x$

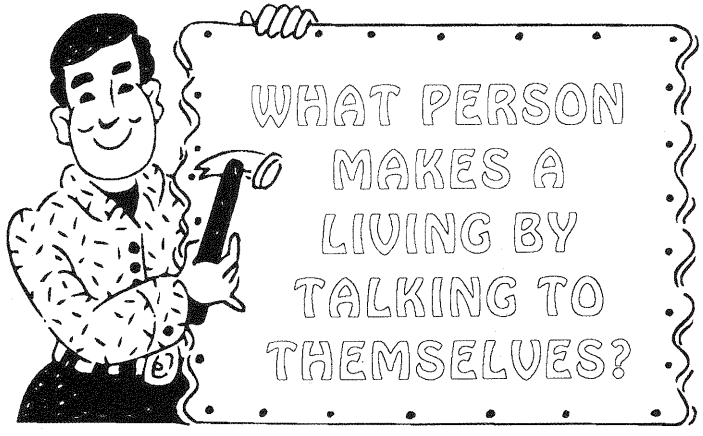
$10 + 2(x + 4) = -5x - 10 + 2x - 7$

$5(6 - 3x) + 9x = 3x - 2(2x + 5) - 20$

$x = -5$ A	$x = -8$ N	$x = \frac{1}{4}$ U	$x = 1$ D	$x = 6$ T
$x = 4$ D	$x = 12$ S	$x = 3$ G	$x = -\frac{1}{2}$ E	$x = 9$ R
$x = 11$ C	$x = \frac{1}{2}$ K	$x = -4$ H	$x = 10$ E	$x = -9$ H
$x = -1$ U	$x = 12$ A	$x = -7$ R	$x = 5$ O	$x = -2$ S

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Solve each of the following equations and look for each solution at the bottom of the page. Write the letter of each equation above its solution.

L $3x + 4 + 2x = 31 - 4x$

S $2(w-2) + 3 = 3(w+1) - 2$

E $-3(m+3) - 2m = 2(6-5m) + 4$

U $\frac{y-3}{4} = y$

T $\frac{x}{4} = x + 3$

O $\frac{2m-3}{3} = m - 3$

I $3x - 2(x+1) = -3(x-1) - 1$

V $4j + 2 - (2j+1) = j + \frac{1}{2}$

R $\frac{k}{6} = \frac{3}{4}$

Q $3m - 3 + 4m = 33 - 2m$

N $3(k+1) - 3 = 2(k-2) + 1$

−0·5	5	−3	−4	4·5	1	3	6	4	−1	1	−2	−4

HOW DOES AN INTRUDER GET INTO A HOUSE?



Solve each of the equations below and look for your answers at the bottom of the page. Write the letter for each equation above the solution.

R $\frac{3x}{2} = 2x - 1$

N $\frac{5x}{3} = 2x + 3$

I $\frac{x+3}{2} = x$

D $\frac{1}{2}(3x + 4) = 3x + 5$

O $\frac{1}{3}(2x + 1) = 3x - 2$

U $\frac{x-5}{2} = x$

W $x - \frac{1}{2} = \frac{3x}{2}$

E $\frac{x-8}{3} = x$

T $\frac{2x-3}{3} = x$

3	-9	-3	2	-5	-2	-4	2	-1	3	-9	-2	1	-1

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WHEN IT RAINS CATS AND DOGS, WHAT DO YOU STEP INTO?



INSTRUCTIONS:
Solve each pair of simultaneous equations. Look for your solution in the boxes below. Cross out each box containing a correct solution. When you finish there will be 7 boxes remaining. Rearrange the boxes according to numerical order to give the answer to the riddle.

$$\begin{aligned} x + y &= 7 \\ 2x + 3y &= 18 \end{aligned}$$

$$\begin{aligned} x + y &= 4 \\ 3x + 2y &= 7 \end{aligned}$$

$$\begin{aligned} 2x + y &= -2 \\ 3x + 2y &= -5 \end{aligned}$$

$$\begin{aligned} -3x + 2y &= 11 \\ 2x + 3y &= -3 \end{aligned}$$

$$\begin{aligned} x + y &= 3 \\ 2x + 4y &= 8 \end{aligned}$$

$$\begin{aligned} x - y &= 1 \\ 4x + 2y &= -14 \end{aligned}$$

$$\begin{aligned} 2x - y &= 12 \\ 4x + 3y &= 14 \end{aligned}$$

$$\begin{aligned} 5x + 2y &= -24 \\ -2x - 3y &= 14 \end{aligned}$$



6 – A (5, – 2)	**3 – O** (2, 4)	**7 – R** (3, 4)	**5 – L** (1, 5)	**1 – P** (– 1,– 6)
4 – T (2,1)	**5 – M** (– 3, 1)	**4 – D** (3, – 1)	**7 – S** (0, 3)	**2 – M** (– 1, 5)
1 – S (1, – 4)	**2 – O** (– 2,– 5)	**3 – E** (– 2,– 3)	**5 – D** (– 4,– 2)	**6 – E** (1, – 2)

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To answer this riddle, solve each pair of similtaneous equations. Look for your solution in the Code Key and notice the letter next to it. Write that letter above the question number in the puzzle below.

WHAT DO YOU GET WHEN YOU CROSS A RUBBISH COLLECTOR WITH A GANGSTER?

1. $x + y = 7$
$x - y = -1$

2. $x + y = 1$
$x - y = 3$

3. $x + y = 3$
$x - y = -3$

4. $x + 2y = -3$
$x + 4y = -5$

5. $x - 3y = -14$
$x + 2y = 16$

6. $2x + y = 8$
$3x - y = 17$

7. $5x + y = -13$
$2x + y = -7$

8. $2x + 2y = 10$
$3x - 2y = 15$

9. $4x + 2y = 18$
$x - 2y = 12$

10. $3x + 2y = -2$
$-3x + y = -10$

CODE KEY	
(0, 3)	R
(2, – 4)	A
(5, – 2)	O
(3, 4)	D
(5, 0)	I
(– 1, – 1)	E
(6, – 3)	G
(2, – 1)	N
(– 2, – 3)	M
(4, 6)	Z

6	3	9	10	2	8	5	4	1	9	3	8	7	4	

WHAT HAPPENED TO THE CANNIBALS WHEN THEY ATE A COMEDIAN?

$\overline{(-1, 1)}$ $\overline{(0, 5)}$ $\overline{(1, -4)}$ $\overline{(6, 3)}$ $\overline{(0, 5)}$ $\overline{(-2, 3)}$ $\overline{(1, 3)}$

$\overline{(-2, 3)}$ $\overline{(2, 4)}$ $\overline{(1, -4)}$ $\overline{(-2, 3)}$ $\overline{(-2, -2)}$ $\overline{(-1, 1)}$

$\overline{(3, -2)}$ $\overline{(2, 4)}$ $\overline{(2, 4)}$ $\overline{(-3, 5)}$ $\overline{(5, 1)}$

Solve each of the following pairs of simultaneous equations and look for your solutions in the code above. Write the letter of each question above its solution.

F

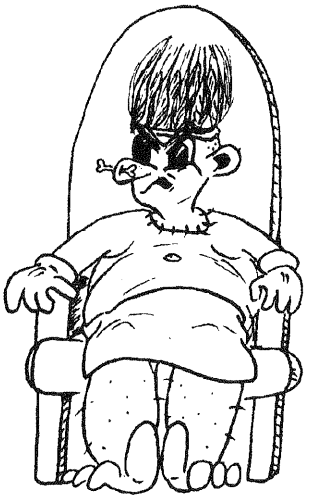
$y = 2x$
$x + y = 6$

N

$y = x - 4$
$x + y = 6$

O

$y = 2x - 8$
$y = x - 5$



T

$y = x + 2$
$x + y = 0$

Y

$y = x - 3$
$x + y = 9$

H

$y = 3x + 5$
$x = y - 5$

E

$y = 3x - 7$
$y = x - 5$

S

$y = x$
$y = 3x + 4$

D

$x = 2y - 5$
$3y + x = 10$

U

$y = 2x + 11$
$y - 3x = 14$

A

$x = 1 - y$
$3y + x = 7$

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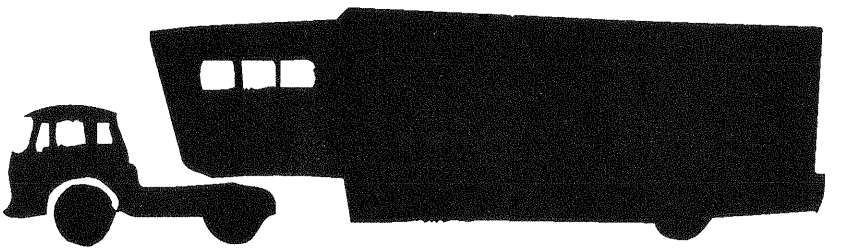
WHY DID IT TAKE FIVE SCOUTS TO HELP THE LITTLE OLD LADY ACROSS THE ROAD?

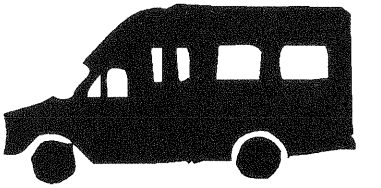
Solve each of the problems below. Look for your answers in the Code Key. Notice the letter next to the correct solution. Write that letter above the number of the question in the code below.

1. The sum of two numbers is 16. Their difference is 8. What are the two numbers?
2. The sum of two numbers is 39. Their difference is 17. What are the two numbers?
3. Three calculators and six pens cost $87. One calculator and four pens cost $33. What is the cost of each calculator and pen?
4. The cost of two books and six magazines is $38. The cost of four books and seven magazines is $56. What is the cost of each book and magazine?
5. Six bags of sugar and two bags of flour weigh 22 kg. Two bags of sugar and five bags of flour weigh 29 kg. How much does each bag of sugar and flour weigh?
6. The sum of two numbers is six and their difference is three. What are the numbers?
7. The sum of John's age and Kerry's age is 19. John is one year younger than three times Kerry's age. How old are John and Kerry?
8. Two bags of apples and three bags of bananas weigh 21·5 kg. Three bags of apples and five bags of bananas weigh 35 kg. How much does each bag of apples and bananas weigh?
9. Four chocolate bars and three icecreams cost $10·20. Two chocolate bars and five icecreams cost $11·40. How much does each chocolate and icecream cost?
10. The sum of two numbers is nine. The larger number is five times the smaller one. What are the two numbers?
11. The sum of two numbers is – 8. One number is three times the other number. What are the two numbers?

Code Key

$7, $4	– H
1·5, 4·5	– T
$8, $6	– M
28, 11	– D
3 kg, 7 kg	– K
14, 5	– I
1·5, 7·5	– G
2·5, 6·5	– B
12, 4	– W
17, 22	– L
$1·20, $1·80	– E
2 kg, 5 kg	– A
7, 2	– R
– 2, – 6	– N
$1·60, $5·80	– Y
$25, $2	– O
9, 7	– F
2·5 kg, 5·5 kg	– S





8	4	9	2	7	2	11	6	1	5	11	6	6	3	10	3	

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MATH O MAGIC 4·4 SOLUTIONS

1. TO GET TO THE SECOND HAND SHOP
2. EVAPORATED MILK
3. HE MELTED
4. WHEN SHE WOKE UP HER PILLOW HAD DISAPPEARED
5. BAKED BEINGS ON TOAST
6. HARESPRAY
7. EX RAY
8. HE BIT HIS NAILS
9. BECAUSE SHE WAS A LIGHT SLEEPER
10. PARACHUTE SCHOOL
11. BECAUSE IT SAID TWIST TO OPEN
12. SO SHE COULD SEE WHEN IT STOPPED RAINING
13. A MISSING PERSONS BUREAU
14. SWALLOW THE LEADER
15. LOCH JAWS
16. THAT KNOCKS AS IT DELIVERS MESSAGES
17. A FUR COAT WITH TOO MANY SLEEVES
18. HE FELL FOR IT
19. ATOMIC ACHE
20. PINS AND NEEDLES
21. UDDER CHAOS
22. VENTRILOQUIST
23. INTRUDER WINDOW
24. POODLES
25. ORGANIZED GRIME
26. THEY HAD A FEAST OF FUN
27. SHE DIDN'T WANT TO GO

